

Interpretable Cascade GNNs for Cyber-Physical Anomaly Detection — Technical Report

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Overview. A temporal graph neural network detects sensor-level cyber-attacks on physics-constrained networks. Part 1 develops and validates the detector on water distribution; Part 2 shows the identical method transfers to a power grid and a road-traffic network. The architecture is an *interpretable cascade*: a fast router ranks a set of attack-specialised experts, and a label-free feedback check accepts or re-routes the choice.

1 Method

Substrate. The network is a graph $G = (V, E)$ with a 6-step sliding window of sparse ($\sim 50\%$ missing), noisy sensor readings. Six experts, each a **GraphSAGE** $\rightarrow$ **GRU** model with reconstruction and per-sensor anomaly heads, are trained under *direct expert supervision* (each expert learns its own attack class regardless of routing) and a mass-conservation penalty $\|B\mathbf{q}\|^2$ (Table 4, 377 K parameters).

Cascade routing. Rather than blending all experts opaquely, the router *rank*s them; the top expert is run and scored by a *label-free feedback* signal

$$s_k = \|\hat{p}_k - p\|_{\text{clean}} + \|B\hat{\mathbf{q}}_k\|^2,$$

(reconstruction consistency on the sensors the expert calls clean, plus a physics residual), standardised across experts. On negative feedback the cascade re-routes to the next-ranked expert, restricted to the router’s top-2 for stability. Every prediction thus carries *which* expert ran and *why* — an audit trail a soft blend cannot provide. Over 5 seeds the cascade reaches $F1\ 0.772 \pm 0.017$, matching the oracle ceiling (0.769) and within a point of a full soft blend (0.789): the interpretability is obtained at negligible accuracy cost.

2 Part 1 — Water distribution (Modena, 272 nodes)

Detection is strong and reproducible (Table 1; $F1\ 0.833$, AUROC 0.979, 5 seeds). Four of the five attack families are detected reliably; **replay** is the exception, and the reasons are made explicit below.

Router diagnostic. Direct measurement confirms a routing failure mode: on Modena the router is healthy (96%), but on the larger Net3 topology it sends **77% of stealthy windows to the replay expert**, because both classes look smooth and 97 nodes give too little context. The cascade’s feedback is designed to contain exactly this case.

Recovering replay. A per-sensor normalisation — rescaling each sensor by its own temporal variance — amplifies the tiny per-node variation a replay lacks. This lifts replay ranking from *below chance* to AUROC 0.79 while overall $F1$ holds (Table 2); it is the single largest contribution to replay.

Why replay still underperforms. Even so, replay $F1$ stays low, and Figure 1 shows why with direct evidence. The anomaly scores of replayed and genuine sensors *overlap* (left): the ranking is informative (AUROC 0.75) but no single threshold separates the two, so the best achievable $F1$ is 0.41 (right). The cause is physical — a replay is a verbatim copy of a past reading, so the only distinguishing cue is the fresh observation noise it lacks. This is an information limit, not a training defect, and it is confirmed by a direct exact-lag-match detector reaching only AUROC 0.62 (missing data and out-of-window lags hide the match).

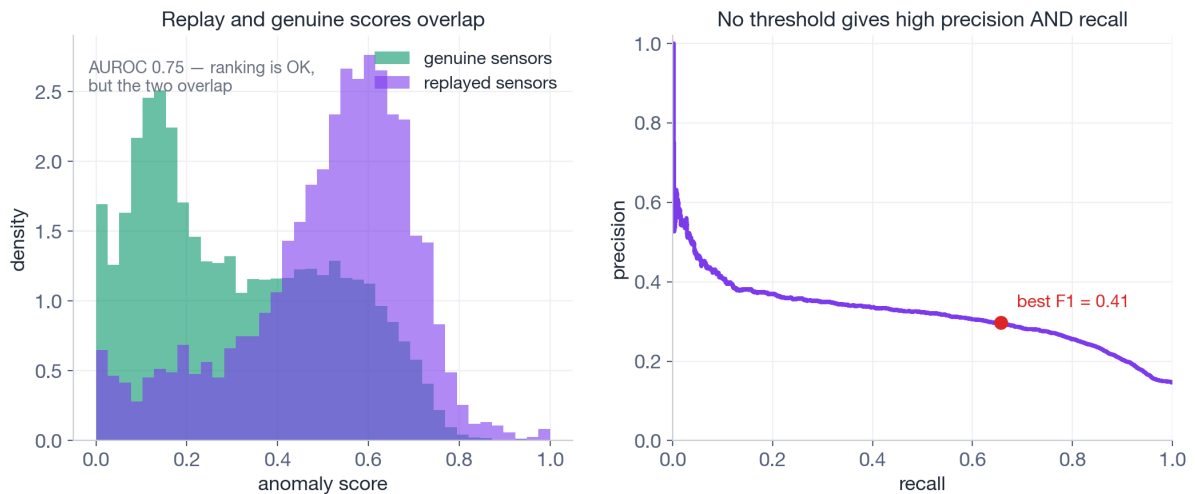


Figure 1: Why replay underperforms. *Left:* replayed (purple) and genuine (green) anomaly scores overlap despite a usable ranking (AUROC 0.75). *Right:* consequently no threshold gives high precision *and* recall; best $F1 = 0.41$.

Table 1: Per-domain results (per-node normalisation, 5 seeds, calibrated threshold). Per-attack columns are F1.

Domain	F1	AUROC	Random	Targeted	Stealthy	Noise	Replay
Water	0.833	0.979	0.989	0.995	0.932	0.912	0.142
Power	0.816	0.959	0.926	0.922	0.664	0.890	0.599
Traffic	0.891	0.980	0.997	0.999	0.665	0.790	0.907

Table 2: Ablation on water (calibrated F1).

Configuration	F1	Replay
Global norm	0.826	0.004
+ per-node norm	0.833	0.142
+ 5-seed ensemble	0.847	0.082

Table 3: Intra-scenario signal variation.

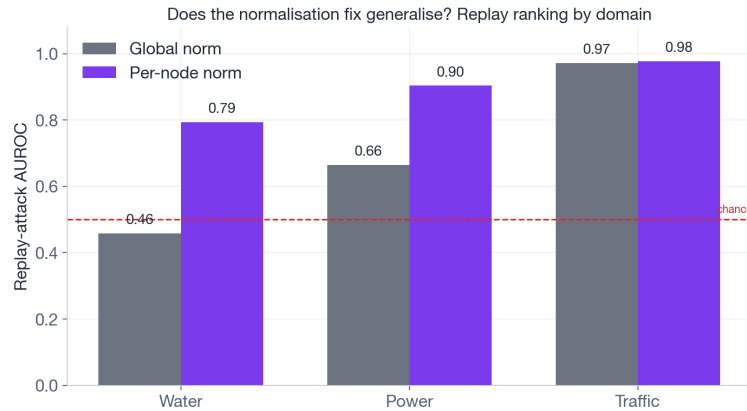
Domain	std	rel.
Water (pressure, m)	5×10^{-6}	10^{-7}
Power (voltage, pu)	7×10^{-4}	10^{-5}
Traffic (speed, km/h)	14.6	1

3 Part 2 — Cross-domain generalisation

The detector, training script and corruption pipeline are kept unchanged; only a per-domain data adapter is added (bus voltage and Kirchhoff’s law for the IEEE 118-bus grid; sensor speed and flow conservation for a 200-sensor road network). Detection quality lands in a tight band across all three physical systems (Table 1; F1 0.82–0.89).

The normalisation fix generalises. Figure 2 shows the replay ranking per domain. The global-normalisation baseline (grey) already tracks *signal speed*: replay is hardest in slow-moving water (0.46) and trivial in fast-moving traffic (0.97) — consistent with the intra-scenario variation spanning seven orders of magnitude (Table 3). Per-node normalisation (purple) lifts replay *most where it is hardest* (water 0.46 → 0.79, power 0.66 → 0.90) and is unnecessary where replay is already easy. The fix is therefore a principled, targeted, cross-domain improvement rather than a water-specific adjustment.

Difficulty inverts. The hard attack class flips by domain: water and power find stealthy easy and replay hard, whereas traffic is the mirror image (Table 1) — a slow drift in traffic speed blends into natural congestion. The domain’s signal dynamics, not the model, determine which attacks are detectable.

**Figure 2:** The normalisation fix generalises. Grey (global norm) tracks signal speed; purple (per-node) lifts replay ranking most where it is hardest. Dashed line = chance.

4 Summary and next steps

An interpretable cascade detector matches the hard-routing ceiling at negligible accuracy cost; a router failure mode is diagnosed and contained; per-node normalisation recovers replay ranking from below chance; and the residual replay gap is shown, with direct evidence, to be an information limit rather than a training defect. All findings transfer unchanged to a power grid and a traffic network, and the normalisation fix generalises in proportion to how hard replay is in each domain. Results are reproducible across 5 seeds per configuration. **Next:** (1) apply the cascade to networks whose experts differ more, so interpretable re-routing also *gains* accuracy; (2) extend the threat model toward the smart-grid FDIA and transport-security literature; (3) consolidate the three domains into a single evaluation and identify a target venue.

Table 4: Model and training configuration.

Backbone	Experts	Hidden (expert/router)	Window	Parameters	Seeds
GraphSAGE+GRU	6	64 / 24	6	376,806	5